

Quadratic B-Spline Averaged Shape Function

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1. Definition

For the static two-phase MPM formulation, the averaged scalar shape function is defined from the same particle-domain average used for cpGIMP:

$$\bar{N}_I(x_p) = \frac{1}{2l_p} \int_{x_p-l_p}^{x_p+l_p} N_I(x) dx.$$

The only difference is the underlying pointwise basis $N_I(x)$. For a one-dimensional quadratic B-spline basis, introduce the non-dimensional coordinate

$$\xi = \frac{x - x_I}{h}, \quad \xi_p = \frac{x_p - x_I}{h}, \quad a = \frac{l_p}{h},$$

where h is grid spacing and x_I is node position. Then

$$\bar{N}_I(x_p) = \frac{1}{2a} \int_{\xi_p-a}^{\xi_p+a} B_2(\xi) d\xi.$$

This is exactly the GIMP averaging definition rewritten for a quadratic B-spline basis.

2. Change of Variables

Starting from the physical-coordinate definition,

$$\bar{N}_I(x_p) = \frac{1}{2l_p} \int_{x_p-l_p}^{x_p+l_p} N_I(x) dx,$$

set

$$x = x_I + h\xi, \quad dx = h d\xi.$$

Then the averaging interval endpoints become

$$\xi^- = \frac{x_p - l_p - x_I}{h} = \xi_p - a, \quad \xi^+ = \frac{x_p + l_p - x_I}{h} = \xi_p + a,$$

so

$$\bar{N}_I(x_p) = \frac{1}{2l_p} \int_{\xi_p-a}^{\xi_p+a} B_2(\xi) h d\xi = \frac{1}{2a} \int_{\xi_p-a}^{\xi_p+a} B_2(\xi) d\xi.$$

3. Interior Quadratic B-Spline

For the interior quadratic B-spline used in the implementation,

$$B_2(\xi) = \begin{cases} \frac{1}{2}\xi^2 + \frac{3}{2}\xi + \frac{9}{8}, & -\frac{3}{2} \leq \xi < -\frac{1}{2}, \\ \frac{3}{4} - \xi^2, & -\frac{1}{2} \leq \xi < \frac{1}{2}, \\ \frac{1}{2}\xi^2 - \frac{3}{2}\xi + \frac{9}{8}, & \frac{1}{2} \leq \xi < \frac{3}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

4. Piecewise Antiderivatives

On the three nonzero subintervals, antiderivatives are

$$F_1(\xi) = \frac{\xi^3}{6} + \frac{3}{4}\xi^2 + \frac{9}{8}\xi,$$

$$F_2(\xi) = \frac{3}{4}\xi - \frac{\xi^3}{3},$$

$$F_3(\xi) = \frac{\xi^3}{6} - \frac{3}{4}\xi^2 + \frac{9}{8}\xi.$$

Therefore the exact averaged shape value is

$$\bar{N}_I(x_p) = \frac{1}{2a} \left(\int_{\xi_p-a}^{\xi_p+a} B_2(\xi) d\xi \right),$$

with the integral evaluated by clipping the averaging interval against the three support pieces:

$$\int_{\xi_p-a}^{\xi_p+a} B_2(\xi) d\xi = [F_1(b_1) - F_1(a_1)] + [F_2(b_2) - F_2(a_2)] + [F_3(b_3) - F_3(a_3)],$$

where

$$\begin{aligned} a_1 &= \max(\xi_p - a, -\tfrac{3}{2}), & b_1 &= \min(\xi_p + a, -\tfrac{1}{2}), \\ a_2 &= \max(\xi_p - a, -\tfrac{1}{2}), & b_2 &= \min(\xi_p + a, \tfrac{1}{2}), \\ a_3 &= \max(\xi_p - a, \tfrac{1}{2}), & b_3 &= \min(\xi_p + a, \tfrac{3}{2}), \end{aligned}$$

and any segment with $b_k \leq a_k$ contributes zero.

5. Explicit Piecewise Evaluation Pattern

The exact average is therefore obtained by the following algorithmic pattern:

$$\bar{N}_I(x_p) = \frac{1}{2a} \sum_{m=1}^3 \left[F_m\left(\min(\xi^+, \xi_m^R)\right) - F_m\left(\max(\xi^-, \xi_m^L)\right) \right]_+,$$

where $[\cdot]_+$ means the contribution is zero if the clipped interval is empty, and

$$(\xi_1^L, \xi_1^R) = (-\tfrac{3}{2}, -\tfrac{1}{2}), \quad (\xi_2^L, \xi_2^R) = (-\tfrac{1}{2}, \tfrac{1}{2}), \quad (\xi_3^L, \xi_3^R) = (\tfrac{1}{2}, \tfrac{3}{2}).$$

This form is preferable in code because it avoids writing many separate cases for different particle sizes or particle-node offsets; the clipping itself handles all intersection patterns.

6. Relation to the Implementation

The implementation in `src/igampm/MPM/static_twophase_ulmpm.py` uses exactly this formula for the `bspline` branch:

- `shape` and `dshape` come from the standard quadratic B-spline basis and gradient;
- `shape_avg` is computed from the exact particle-domain average above;
- this removes the previous mismatch where the `bspline` branch reused the GIMP averaging rule.

7. Remarks

This derivation addresses only the averaged scalar weight $\bar{N}_I$ needed by the pressure projection and PPP stabilization terms. The global residual and Hessian assembly remain unchanged once the correct averaged weights are supplied.